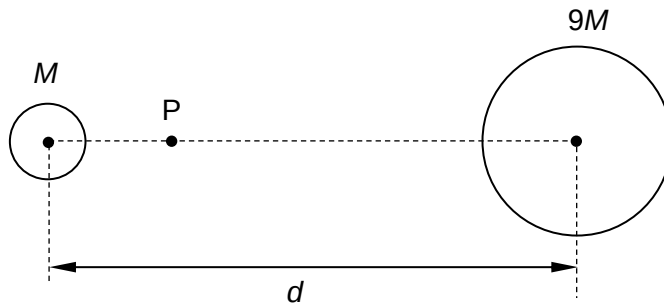


10

Two isolated stars, of mass M and $9M$, are separated in space by a distance d between the centres of the stars.



Point P is a point, between the centres of the stars, where the gravitational field strength is zero.

What is the gravitational potential at P ?

A

$$-\frac{10GM^2}{d}$$

B

$$-\frac{16GM}{d}$$

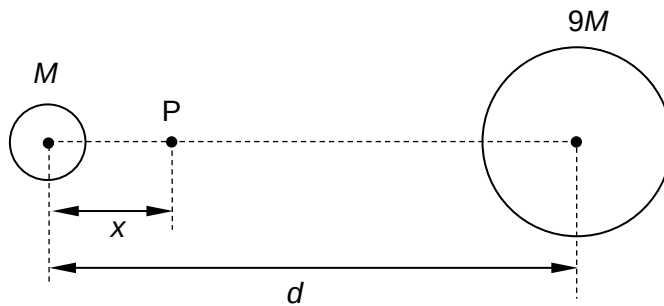
C

$$-\frac{20GM^2}{d}$$

D

$$-\frac{32GM}{d}$$

Answer: B



x – distance from the centre of star mass M .

At P , where gravitational field strength is zero,

$$\frac{GM}{x^2} = \frac{G(9M)}{(d-x)^2}$$

$$(d-x)^2 = 9x^2$$

$$d-x = 3x$$

$$x = \frac{1}{4}d$$

$$\text{Gravitational potential at } P = -\frac{GM}{\frac{1}{4}d} - \frac{G(9M)}{\frac{3}{4}d} = -\frac{GM}{d} \left(4 + \frac{4}{3} \times 9 \right) = -\frac{16GM}{d}$$

11	The International Space Station (ISS) orbits the Earth with uniform speed above the Earth's
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